

Ramnik Lidhar

Brampton, ON | (437) 255-5267 | ramniklidhar@gmail.com | ramnikl.github.io

SUMMARY OF QUALIFICATIONS

- Third-year Electrical Engineering student at Toronto Metropolitan University with hands-on experience in embedded systems, firmware development, and FPGA-based hardware design
- Designed, implemented, and debugged an 8-bit general-purpose processor (ALU, FSM control unit, and decoder) in VHDL using Quartus II, validating functionality through simulation and waveform analysis
- Built an ESP32-based embedded firmware project in C/C++, integrating I2C peripherals and debugging hardware/firmware interaction issues at the driver and communication level
- Experienced in Linux development environments and technical documentation, with strong communication and teamwork skills from collaborative engineering projects

TECHNICAL SKILLS

Programming Languages: C, C++, VHDL, Java, JavaScript

Software & Tools: Quartus II, Git, KiCad, MATLAB, Microsoft Office Suite

Technical Areas: Embedded systems, Firmware Development, FPGA Design, Hardware/Firmware Debugging

Operating Systems: Linux, Windows, Mac

EDUCATION

Bachelor of Engineering (B.Eng.): *Electrical Engineering*

Sept. 2024 - Exp. May 2029

Toronto Metropolitan University (**cGPA: 3.97/4.33**)

- Relevant Coursework: Microprocessors, Digital Systems, Electronic Circuits I

ACADEMIC AND PERSONAL PROJECTS

General-Purpose Processor & ALU Design | Digital Systems

Sept. 2025 - Dec. 2025

- Designed and implemented an 8-bit general-purpose processor in VHDL (Quartus), including dual 8-bit storage registers, a 9-state FSM control unit, a 4-to-16 decoder, and three ALU variants supporting arithmetic, logic, and bitwise rotation/shift operations
- Verified all designs via Quartus simulation and waveform analysis across 9 operation states, debugging timing and logic errors with a project partner

NFC-Based Embedded Alarm Clock | Personal Project

Jul. 2026 - Aug. 2026

- Designed and built an embedded alarm clock on an ESP32, integrating an NFC reader (I2C), RTC module, OLED display, and button-driven menu system in C/C++
- Debugged hardware/firmware interaction issues including driver conflicts and I2C communication faults
- Implemented NFC-based alarm dismissal requiring a physical card/tag scan, with RTC-driven once-daily alarm triggering

Multi-Stage MOSFET Amplifier Design | Electronic Circuits I

Jan. 2026 - Apr. 2026

- Designed a 4-stage common-source MOSFET amplifier in SPICE, achieving 61.1 dB voltage gain (exceeding the 60 dB spec) while keeping power consumption under 0.89 mW and bandwidth above the required 500 kHz
- Performed hand calculations for transistor sizing, biasing, and gain, then validated the design against SPICE simulation, identifying and explaining sources of discrepancy (up to 37% in drain current) between theoretical and simulated results

ADDITIONAL EXPERIENCE

Self-Hosted Server Deployment | Personal Project

Sept. 2024 - Present

- Deployed and managed a containerized application server (Crafty on CasaOS) on Ubuntu Linux, handling networking, backups, and multi-user access control
- Diagnosed and resolved container/service-level issues to maintain uptime